SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Bhavatharani H (192126025b) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Bhavani .

Criteria	Weightage	Q1 ~~10~~ 8marks	Q2 ~~10~~ 8marks	Q3 ~~10~~ 8marks	Q4 ~~10~~ 8marks	Q5 8marks
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	2	2	2	2
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	1	1	1	0	1
Interpretation of Results	15%	1	0	1	0	0

Total Marks 36 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING

Assessment Tool 1 (AT1) - Analytical Problem Solving

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Register No.	1921260256
Course Code	CSA0709
Course Name	Computer Networks
Course Outcome	CO2 - Configure IP addressing schemes, subnetting and basic routing techniques using a simulation tool, for efficient network design and topology (BL3)
Total	5 questions x 8 marks = 40 marks

Question 1 (8 marks) - Subnetting a Class C network (FLSM)

An organisation is allotted the network **192.168.10.0/24**. It needs **6 equal-sized subnets** for six labs. Determine the new subnet mask, number of subnets and hosts per subnet, and tabulate the network address, usable host range and broadcast address of each required subnet.

Answer

Step 1 - Borrow bits (2 marks). Subnets needed = 6, so 2^n >= 6 gives n = 3 (2^3 = 8 subnets, 2 spare). The mask goes from /24 to **/27**.

Step 2 - Mask and hosts (2 marks). /27 = 255.255.255.224. Host bits = 5, so hosts per subnet = 2^5 - 2 = **30**. Block size = 256 - 224 = **32**.

Step 3 - Subnet table (3 marks).

Subnet	Network	First host	Last host	Broadcast
1	192.168.10.0	192.168.10.1	192.168.10.30	192.168.10.31
2	192.168.10.32	192.168.10.33	192.168.10.62	192.168.10.63
3	192.168.10.64	192.168.10.65	192.168.10.94	192.168.10.95
4	192.168.10.96	192.168.10.97	192.168.10.126	192.168.10.127
5	192.168.10.128	192.168.10.129	192.168.10.158	192.168.10.159
6	192.168.10.160	192.168.10.161	192.168.10.190	192.168.10.191

Step 4 - Analysis (1 mark). Subnets 7 (192.168.10.192/27) and 8 (192.168.10.224/27) remain free for future growth. Total usable hosts = 6 x 30 = 180, versus 254 in the undivided network; the loss (network and broadcast addresses per subnet) is the price of segmentation, which gives smaller broadcast domains and better security.

Question 2 (8 marks) - Analysing a host address in a Class B network (CIDR)

A host has the address **172.16.45.200/20**. Find (a) the subnet mask, (b) the network address, (c) the broadcast address, (d) the usable host range, (e) the number of usable hosts, and (f) whether the host 172.16.50.10 belongs to the same subnet.

Answer

(a) Mask (1 mark). /20 = 11111111.11111111.**1111**0000.00000000 = **255.255.240.0**.

(b) Network address (2 marks). Third octet: 45 = 00101101; mask 240 = 11110000. AND gives 00100000 = 32. Network = **172.16.32.0/20**.

(c) Broadcast (2 marks). Set all 12 host bits to 1: third octet = 0010**1111** = 47, fourth = 255. Broadcast = **172.16.47.255**.

(d) Host range (1 mark). 172.16.32.1 to 172.16.47.254.

(e) Usable hosts (1 mark). $2^{12} - 2$ = **4094**.

(f) Same subnet? (1 mark). 172.16.50.10 has third octet 50, which is greater than 47, so it lies in the next block (172.16.48.0/20). It is **not** in the same subnet, and a router is required for the two hosts to communicate.

Question 3 (8 marks) - Variable Length Subnet Masking (VLSM) design

A company has the block **192.168.1.0/24** and the following requirements: Dept A - 100 hosts, Dept B - 50 hosts, Dept C - 25 hosts, Dept D - 10 hosts, and one router-to-router serial link (2 hosts). Design an efficient addressing plan using VLSM and show that fixed-length subnetting cannot satisfy it.

Answer

Method (2 marks). Sort requirements from largest to smallest and allocate each from the next free address, using the smallest block that fits (2^h - 2 >= hosts).

Allocation (4 marks).

Segment	Hosts	Block / Mask	Network	Usable range	Broadcast
Dept A	100	/25 (128) 255.255.255.128	192.168.1.0	.1 - .126	192.168.1.127
Dept B	50	/26 (64) 255.255.255.192	192.168.1.128	.129 - .190	192.168.1.191
Dept C	25	/27 (32) 255.255.255.224	192.168.1.192	.193 - .222	192.168.1.223
Dept D	10	/28 (16) 255.255.255.240	192.168.1.224	.225 - .238	192.168.1.239
WAN link	2	/30 (4) 255.255.255.252	192.168.1.240	.241 - .242	192.168.1.243

Analysis (2 marks). Addresses used = 128 + 64 + 32 + 16 + 4 = 244 of 256, leaving 192.168.1.244 - .255 (12 addresses) free. With FLSM, the largest department (100 hosts) forces every subnet to /25, which yields only 2 subnets in a /24, so five segments cannot be fitted. VLSM therefore saves address space and supports the design.

Question 4 (8 marks) - Static routing configuration and verification

Three routers are connected in a line: R1 - R2 - R3. LANs: R1 = 192.168.1.0/24, R2 = 192.168.2.0/24, R3 = 192.168.3.0/24. Links: R1-R2 = 10.0.0.0/30 (R1 .1, R2 .2); R2-R3 = 10.0.0.4/30 (R2 .5, R3 .6). Write the static routes needed for full connectivity, suggest an optimisation, and state how you would verify it in a simulator such as Cisco Packet Tracer.

Answer

Analysis (2 marks). Each router knows only its directly connected networks. R1 must learn 192.168.2.0, 192.168.3.0 and 10.0.0.4/30; R2 must learn 192.168.1.0 and 192.168.3.0; R3 must learn 192.168.1.0, 192.168.2.0 and 10.0.0.0/30. Next hop is always the neighbour router on the shared link.

Configuration (3 marks).

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
R1(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.2
R1(config)# ip route 10.0.0.4   255.255.255.252 10.0.0.2

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.0.0.6

R3(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.5
R3(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.5
R3(config)# ip route 10.0.0.0   255.255.255.252 10.0.0.5
```

Optimisation (1.5 marks). R1 and R3 are stub routers, so each can use a single default route instead: `ip route 0.0.0.0 0.0.0.0 10.0.0.2` on R1 and `ip route 0.0.0.0 0.0.0.0 10.0.0.5` on R3. This reduces the routing table to one entry per stub router.

Verification (1.5 marks). Use `show ip route` (routes appear with code 'S'), then `ping 192.168.3.10` from a PC in 192.168.1.0 and `tracert` to confirm the path R1 - R2 - R3. Limitation: static routes do not adapt to link failure and become hard to maintain as the network grows, which motivates dynamic routing (Question 5).

Question 5 (8 marks) - Dynamic routing: RIP v2 vs OSPF

For the same topology as Question 4, the R1-R2 link is FastEthernet (100 Mbps), the R2-R3 link is a serial link of 1.544 Mbps, and the R3 LAN is FastEthernet. (a) Configure RIPv2 and OSPF on R2. (b) Compute the RIP metric and the OSPF cost from R1 to 192.168.3.0/24. (c) Compare the two protocols and justify a choice.

Answer

(a) Configuration on R2 (3 marks).

```
! RIP version 2
R2(config)# router rip
R2(config-router)# version 2
R2(config-router)# no auto-summary
R2(config-router)# network 192.168.2.0
R2(config-router)# network 10.0.0.0

! OSPF (single area 0)
R2(config)# router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.0.0.0   0.0.0.3   area 0
R2(config-router)# network 10.0.0.4   0.0.0.3   area 0
```

(b) Metrics (2.5 marks). RIP uses hop count: R1 to R2 to R3 gives metric **2** to 192.168.3.0 (maximum usable is 15; 16 means unreachable). **OSPF** cost = reference bandwidth / interface bandwidth, with the default reference of 100 Mbps. FastEthernet = 100/100 = 1; serial 1.544 Mbps = 100,000,000 / 1,544,000 = 64.77, truncated to 64. Cost from R1 = 1 (R1-R2) + 64 (R2-R3) + 1 (R3 LAN) = **66**.

(c) Comparison (2.5 marks).

Feature	RIPv2	OSPF
Type	Distance vector	Link state
Metric	Hop count (ignores bandwidth)	Cost based on bandwidth
Convergence	Slow (periodic updates every 30 s)	Fast (event-triggered LSAs)
Scalability	Limited to 15 hops	Large networks, area hierarchy
Administrative distance	120	110
Overhead / complexity	Low, simple to configure	Higher CPU/memory, more complex

Justification. RIP would treat the slow serial link and a fast link as equal (1 hop), so it can choose a poorer path. OSPF accounts for bandwidth and converges faster, so it is the better choice for anything beyond a small network, while RIPv2 suffices for a small, simple lab topology.

Marks summary: Q1 8 + Q2 8 + Q3 8 + Q4 8 + Q5 8 = 40 marks.